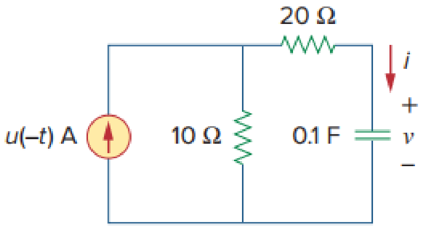
Chapter 7, Problem 48P

Problem

Find $v(t)$ and $i(t)$ in the circuit of Fig. 7.115.



Step-by-step solution

Step 1 of 6

It is known that,

$$u(-t) = \begin{cases} 1 & t < 0 \\ 0 & t \geq 0 \end{cases}$$

Refer to circuit diagram in Figure 7.115 in the textbook.

For $t < 0$ the value of current source is 1 A and circuit reaches steady state. Under steady state capacitor acts as open circuit. The modified circuit is shown in Figure 1.

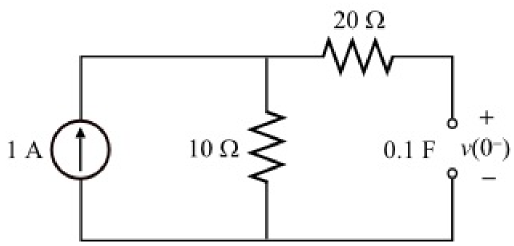


Figure 1

Step 2 of 6

The capacitor in the circuit is open circuit; no current is flowing through $20\ \Omega$ resistor. The voltage across the $10\ \Omega$ resistor is the voltage across the capacitor.

Calculate the voltage across the $10\ \Omega$ resistor at $t = 0^-$.

$$\begin{aligned} v_{10\Omega} &= (10\ \Omega)(1\ \text{A}) \\ &= 10\ \text{V} \end{aligned}$$

Calculate the initial voltage across the capacitor at $t = 0^-$.

$$\begin{aligned} v(0^-) &= v_{10\Omega} \\ &= 10\ \text{V} \end{aligned}$$

The voltage across the capacitor doesn't change instantly.

$$\begin{aligned} v(0^-) &= v(0^+) \\ &= 10\ \text{V} \end{aligned}$$

Step 3 of 6

For $t=0$ the current source $\left(u(-t)=0 \text{ for } t \geq 0\right)$ is open circuit. The modified circuit is shown in Figure 2.

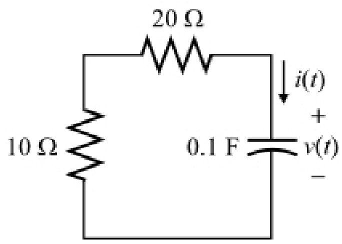


Figure 2

Step 4 of 6

Circuit shown in Figure 2 is a source free RC circuit.

Write the expression for voltage across the capacitor in a source free RC circuit.

$$v(t)=v(0^+)e^{\frac{-t}{\tau}}$$

Here,

τ is the time constant.

Calculate the time constant.

$$\begin{aligned}\tau &= R_{eq}C \\ &= (10\ \Omega + 20\ \Omega)(0.1\ \text{F}) \\ &= (30\ \Omega)(0.1\ \text{F}) \\ &= 3\ \text{s}\end{aligned}$$

Step 5 of 6

Calculate the voltage across the capacitor for $t \geq 0$.

$$v(t)=v(0^+)e^{\frac{-t}{\tau}}$$

Substitute $10\ \text{V}$ for $v(0^+)$ and $3\ \text{s}$ for τ in the equation.

$$\begin{aligned}v(t) &= (10\ \text{V})e^{\frac{-t}{3}} \\ &= 10e^{-0.333t}\ \text{V}\end{aligned}$$

Therefore, the voltage, $v(t)$ across the capacitor for $t \geq 0$ is $\boxed{10e^{-0.333t}\ \text{V}}$.

Step 6 of 6

Calculate the current through the capacitor.

$$\begin{aligned}i(t) &= C\frac{dv(t)}{dt} \\ &= (0.1\ \text{F})\frac{d}{dt}\left(10e^{-0.333t}\ \text{V}\right) \\ &= (0.1)(-0.333)\left(10e^{-0.333t}\right)\ \text{A} \\ &= -0.333e^{-0.333t}\ \text{A}\end{aligned}$$

Therefore, the current, $i(t)$ through the capacitor for $t \geq 0$ is $\boxed{-0.333e^{-0.333t}\ \text{A}}$.